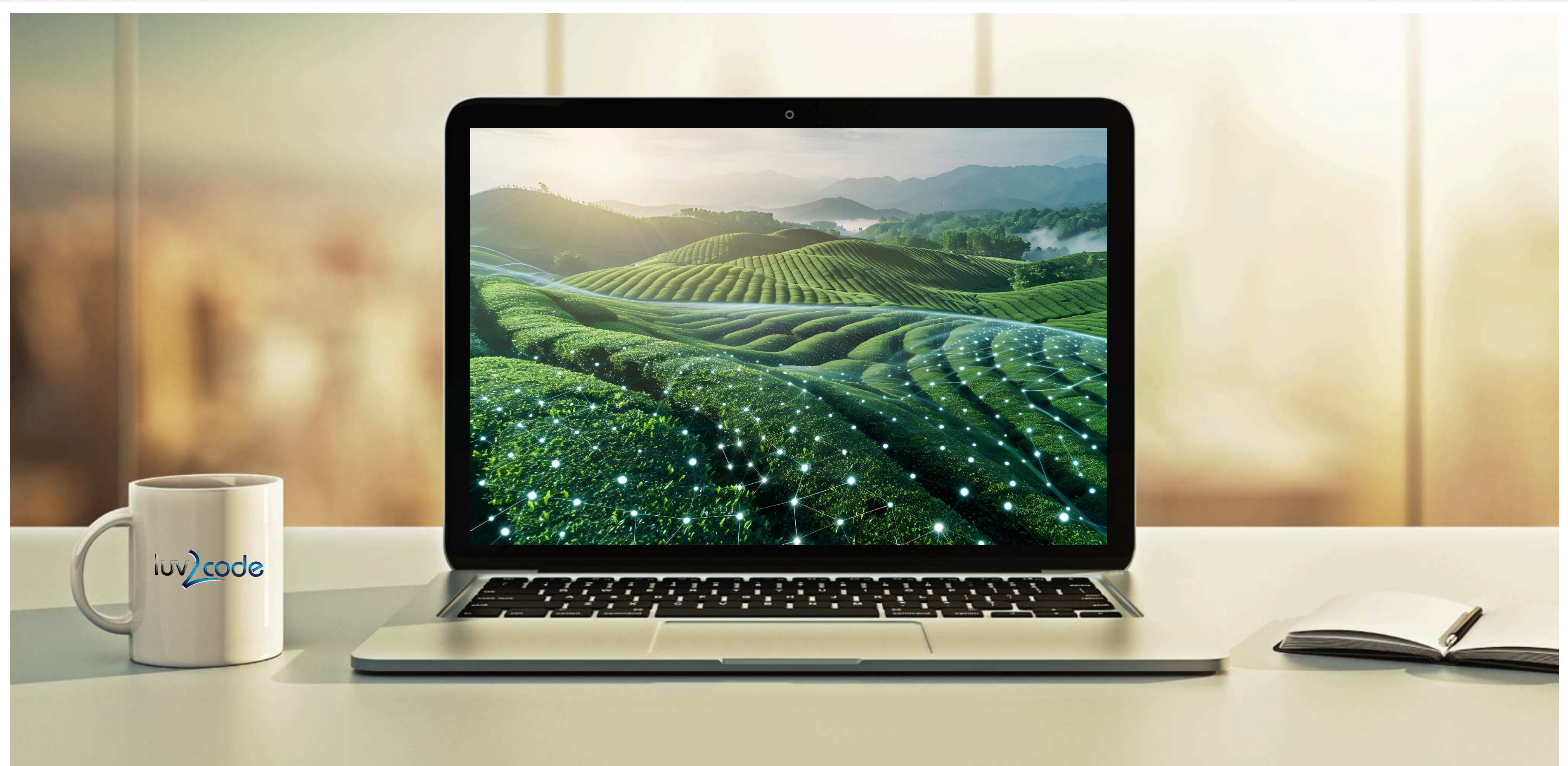


Course Introduction



Course Development Team



Eric Roby

www.codingwithroby.com



Chad Darby

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Course Objective

A lot to cover!

Learn how to build professional Spring Boot REST applications using authentication and databases.

You will learn how to ...

- Develop Spring Boot REST applications
- Leverage Hibernate/JPA for database access
- Develop REST API endpoints using Spring Boot
- Apply Spring Security to control application access.
- Implement and leverage your own JWTs you build from scratch.
- Add Roles and Admin status throughout your REST endpoints
- Leverage all Java configuration (no xml) and Maven

Source Code and PDFs

All source code is available for download

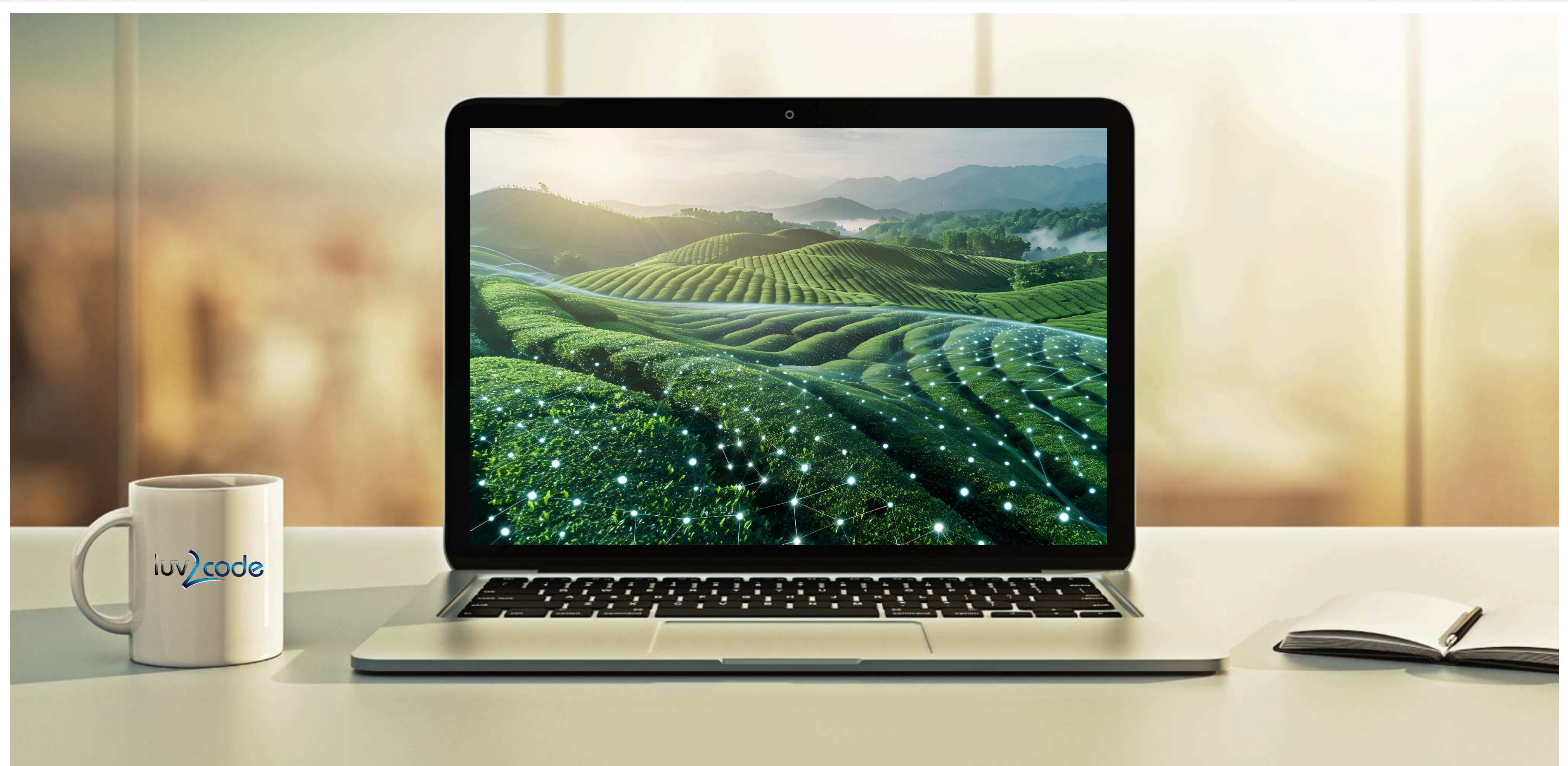
All PDFs of slides are available for download

Questions / Help

If you have questions or need help...

Post the question in the classroom discussion forum

Java Development Environment



REST APIs - REST Web Services



You will learn how to ...

- Create REST APIs / Web Services with Spring
- Discuss REST concepts, JSON and HTTP messaging
- Install REST client tool: Swagger
- Develop REST APIs / Web Services with **@RestController**
- Build a CRUD interface to the database with Spring REST

Practical Results

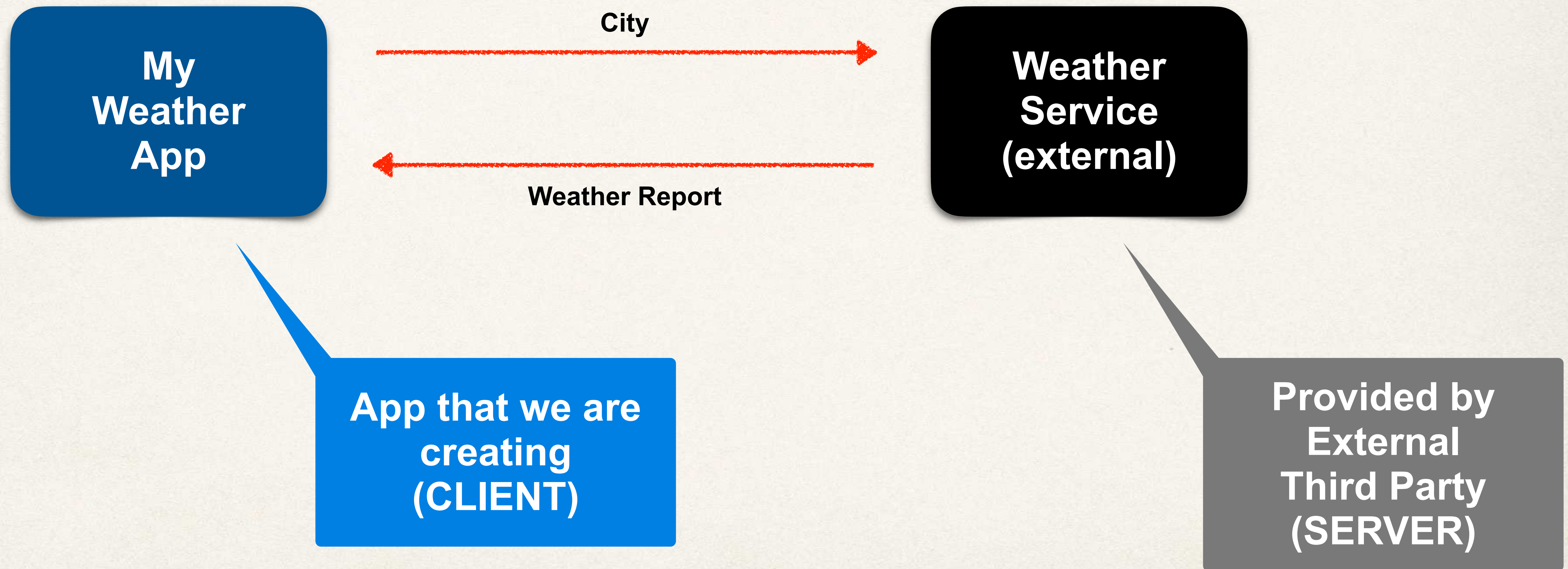
- Introduction to Spring REST development
- Not an A to Z reference ... for that you can see [Spring Reference Manual](#)

<https://projects.spring.io/spring-framework/>

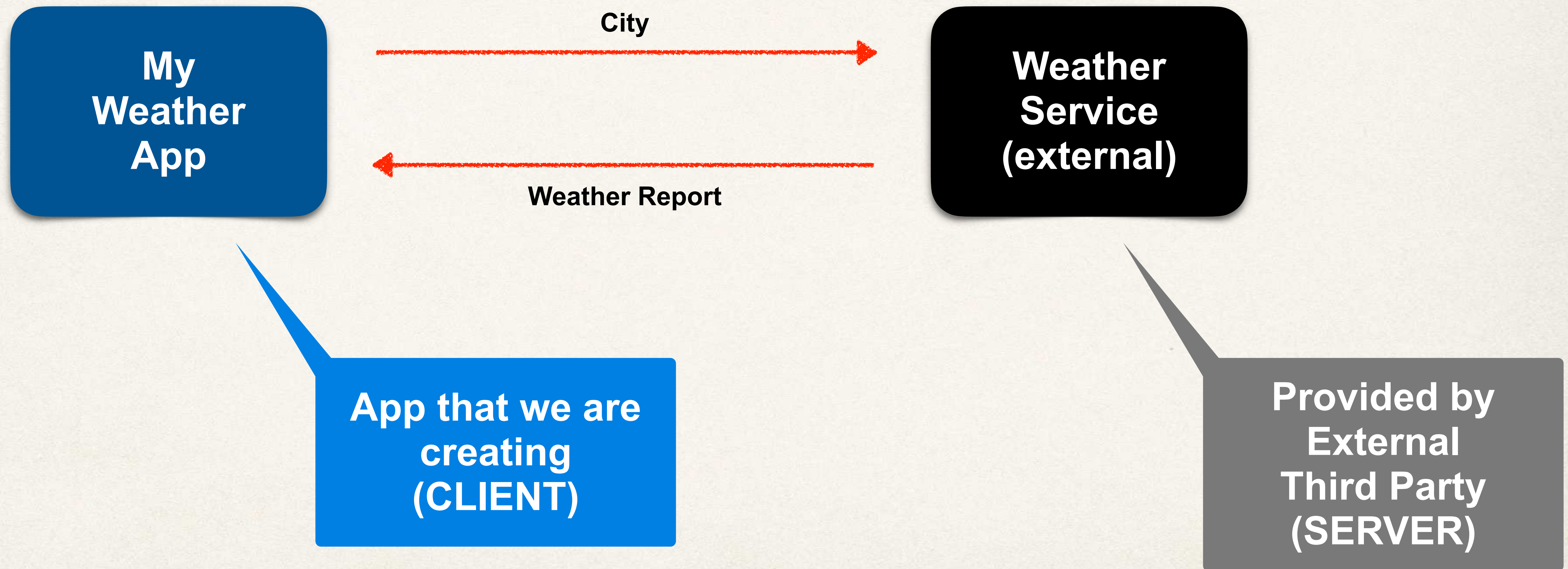
Business Problem

- Build a client app that provides the weather report for a city
- Need to get weather data from an external service

Application Architecture

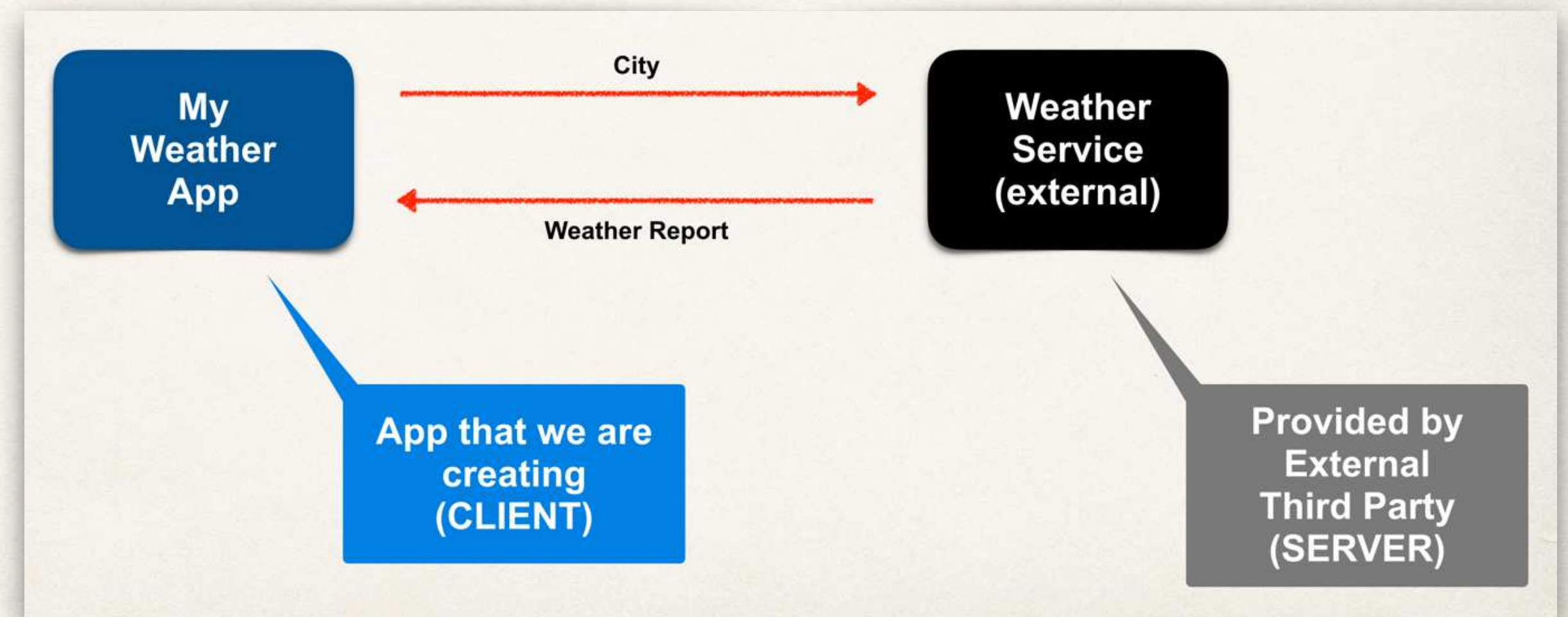


Application Architecture



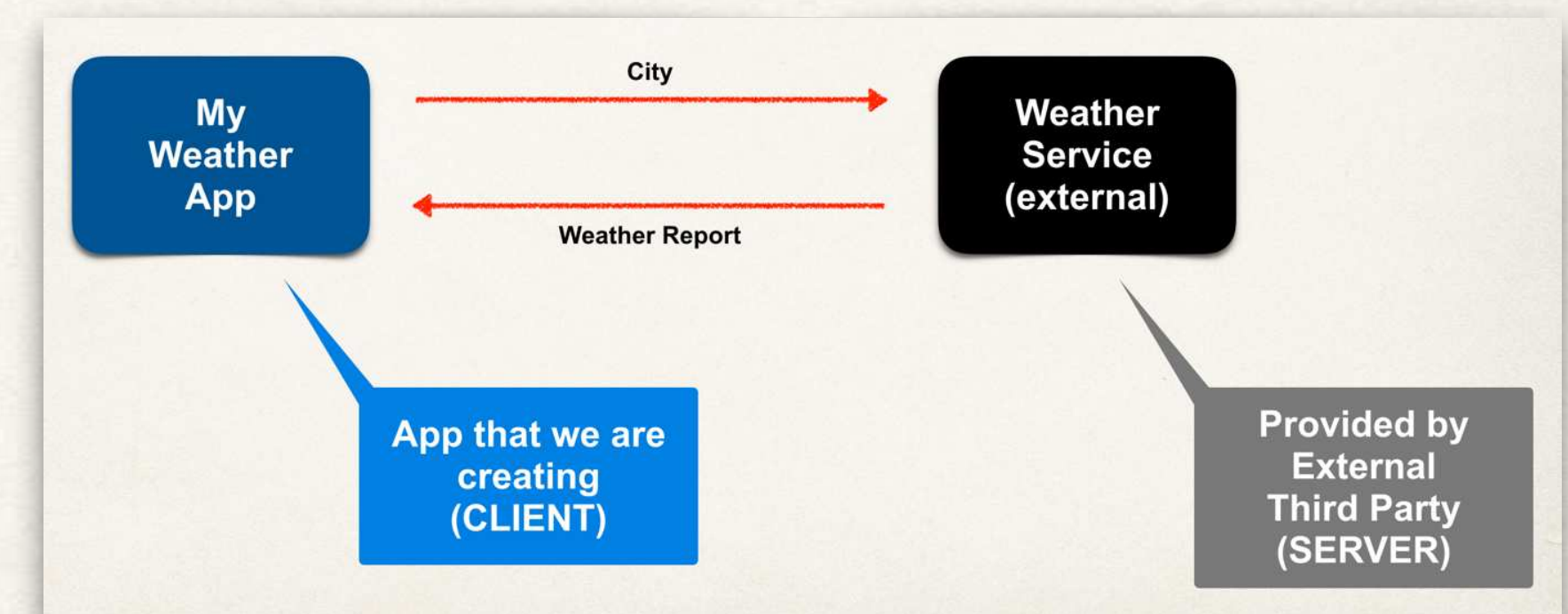
Questions

- How will we connect to the Weather Service?
- What programming language do we use?
- What is the data format?



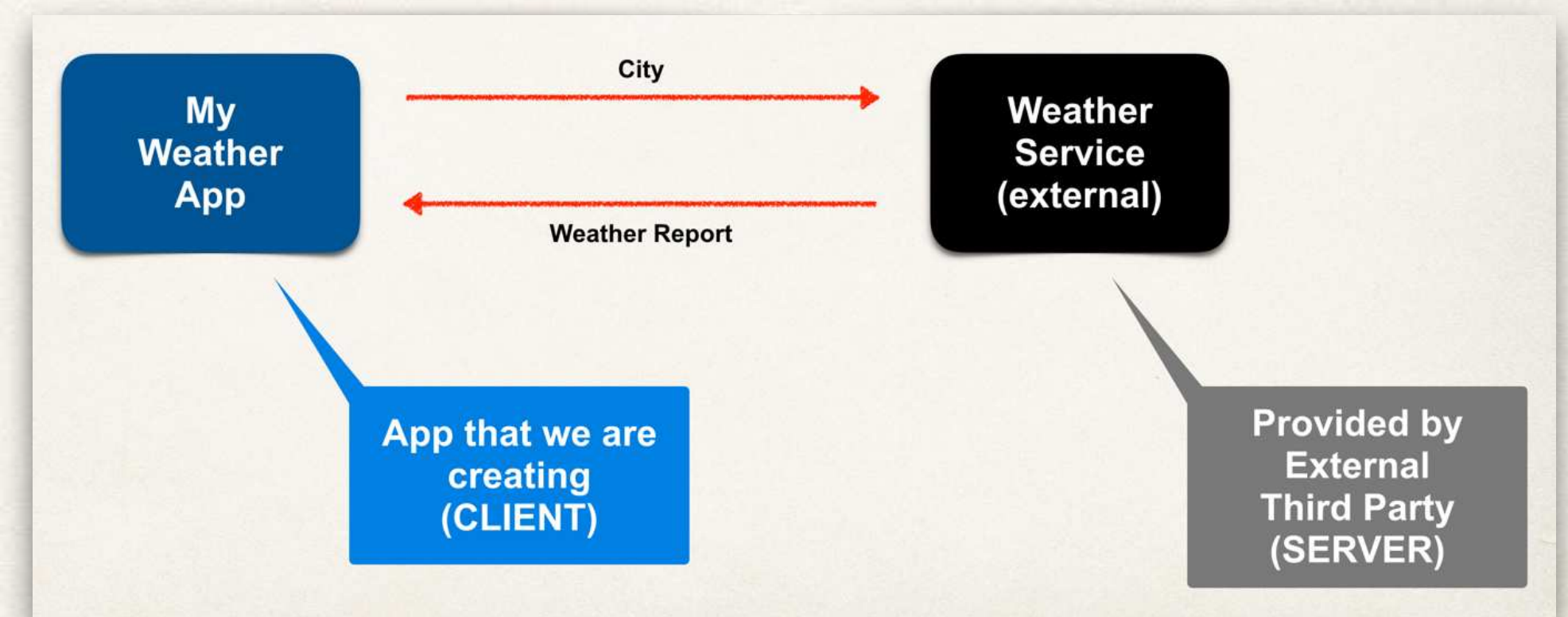
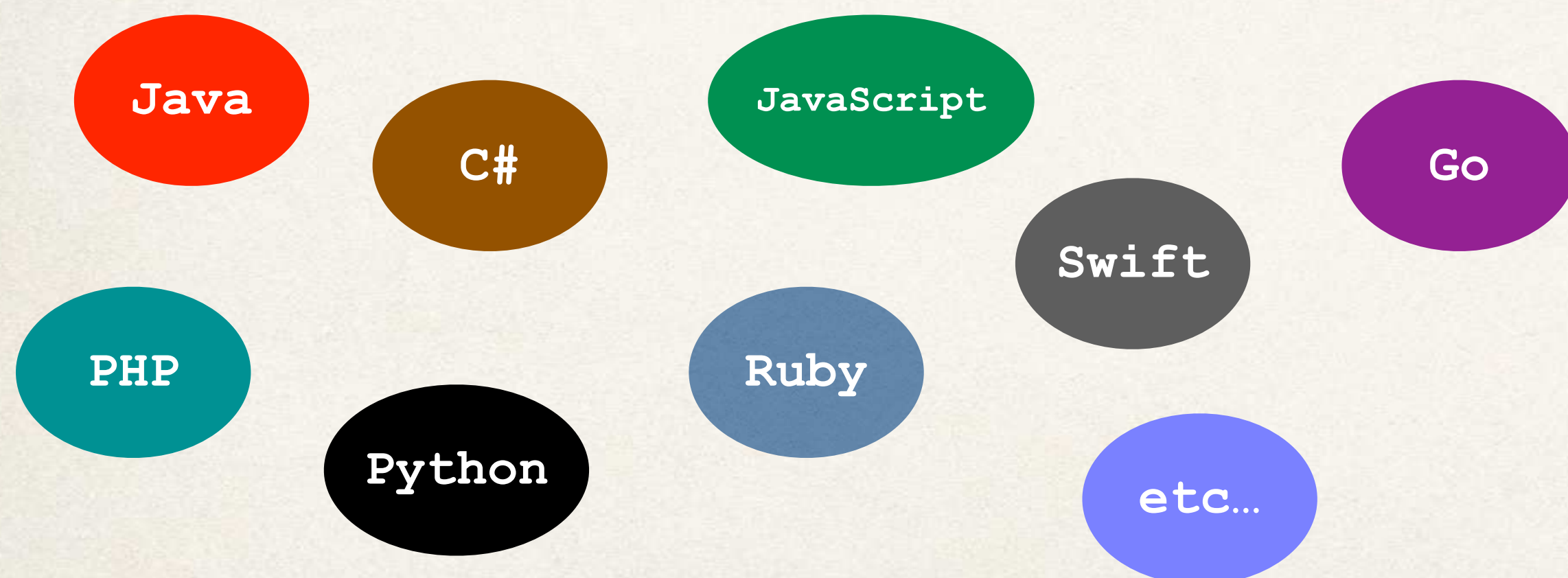
Answers

- How will we connect to the Weather Service?
- We can make REST API calls over HTTP
- REST: REpresentational State Transfer
- Lightweight approach for communicating between applications



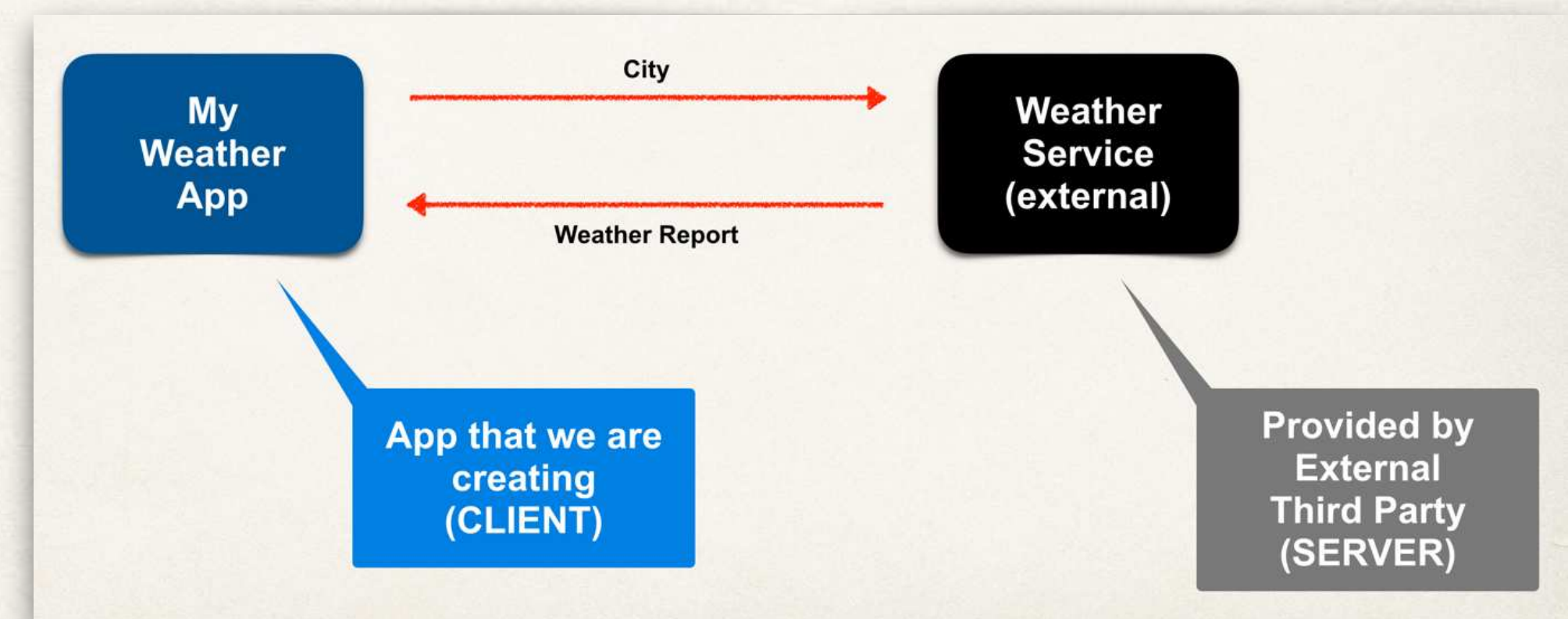
Answers

- What programming language do we use?
- REST is language independent
- The **client** application can use **ANY** programming language
- The **server** application can use **ANY** programming language



Answers

- What is the data format?
 - REST applications can use any data format
 - Commonly see XML and JSON
 - JSON is most popular and modern
 - JavaScript Object Notation

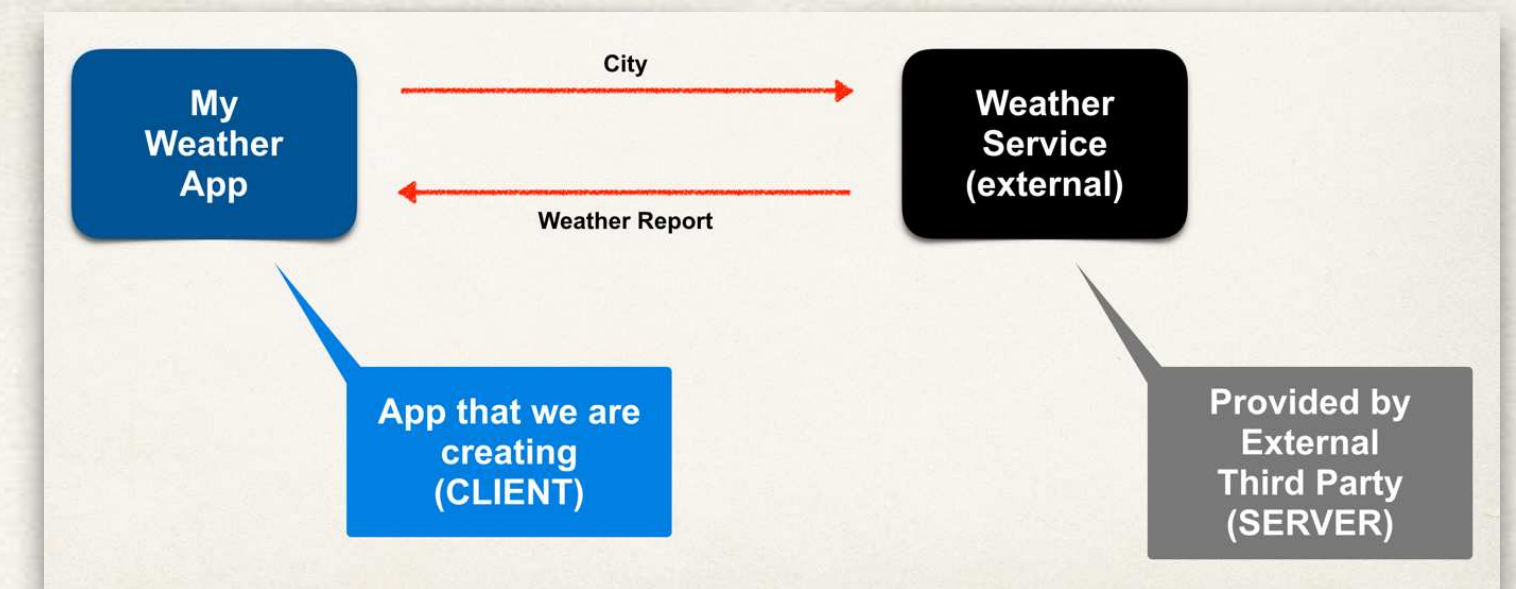


Possible Solution

- Use online Weather Service API provided by: openweathermap.org
- Provide weather data via an API
- Data is available in multiple formats: JSON, XML etc ...

Call Weather Service

- The API documentation gives us the following:
 - Pass in the latitude and longitude for your desired location

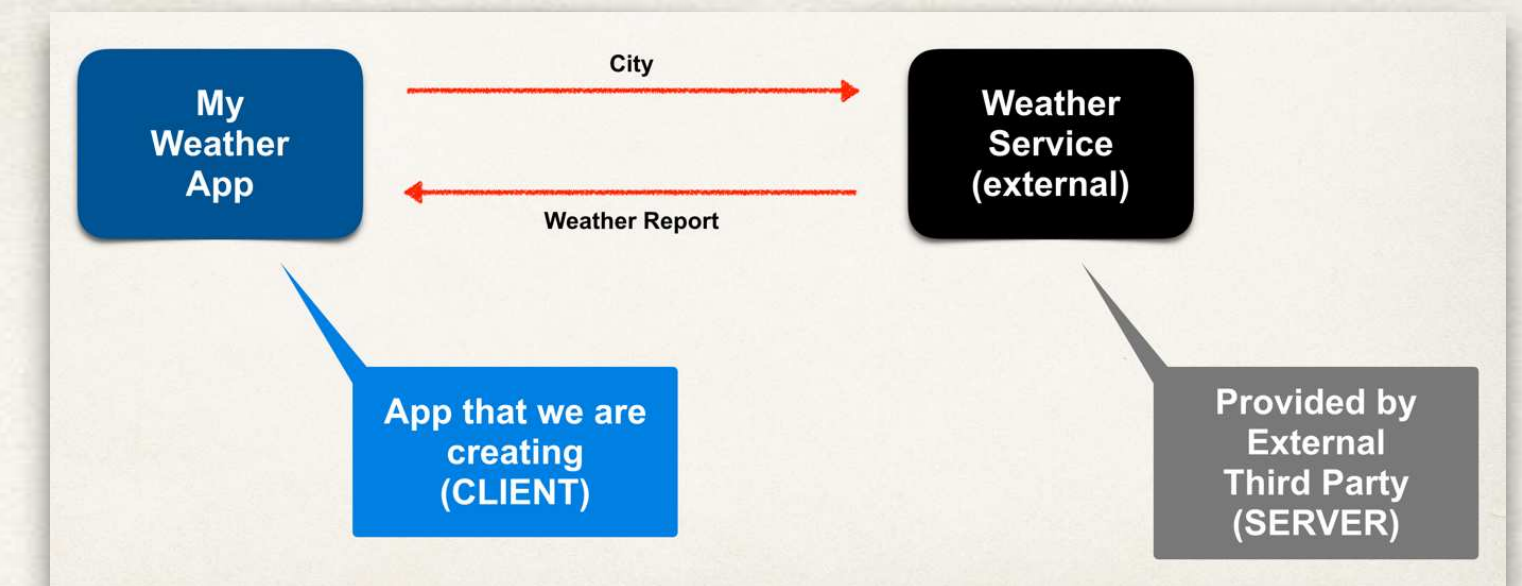


```
api.openweathermap.org/data/<apiVersion>/onecall?lat={theLatitude}&lon={theLongitude}
```

**Pass the
latitude and longitude
of the city**

Response - Weather Report

- The Weather Service responds with JSON



```
{  
  ...  
  "temp": xxx,  
  "feels_like": yyy,  
  "humidity": zzz,  
  ...  
}
```

Condensed
version

Multiple Client Apps

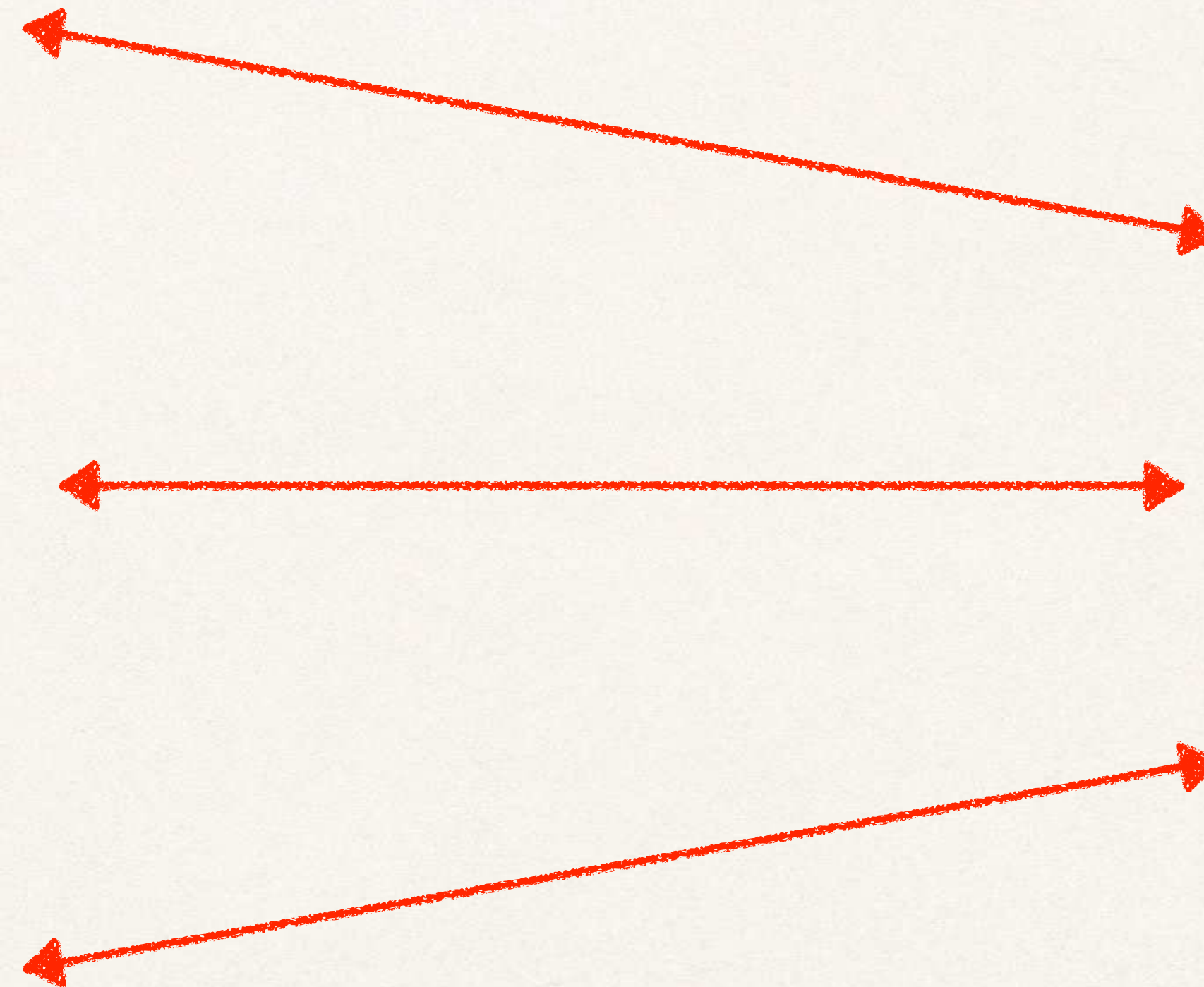
Remember:
REST calls can be made over HTTP
REST is language independent

My
Weather
Spring MVC

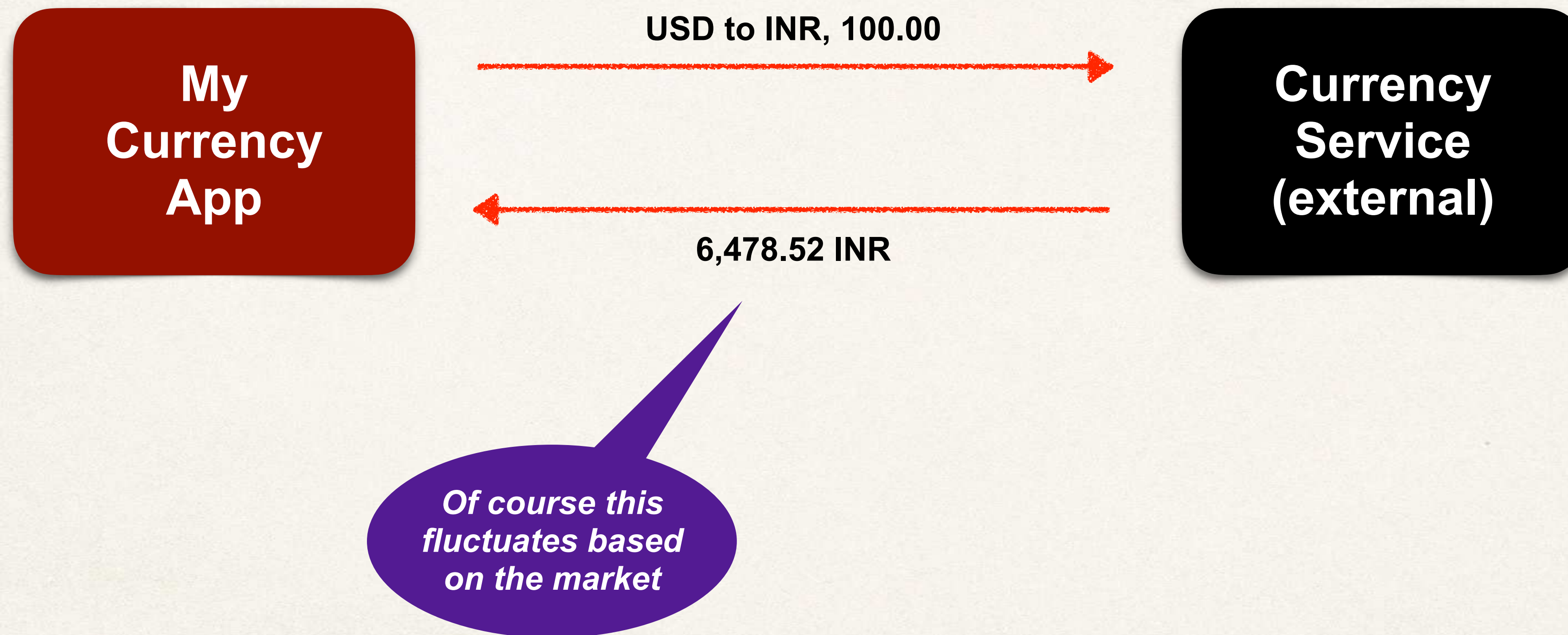
My
Weather
C# App

My
Weather
iPhone App

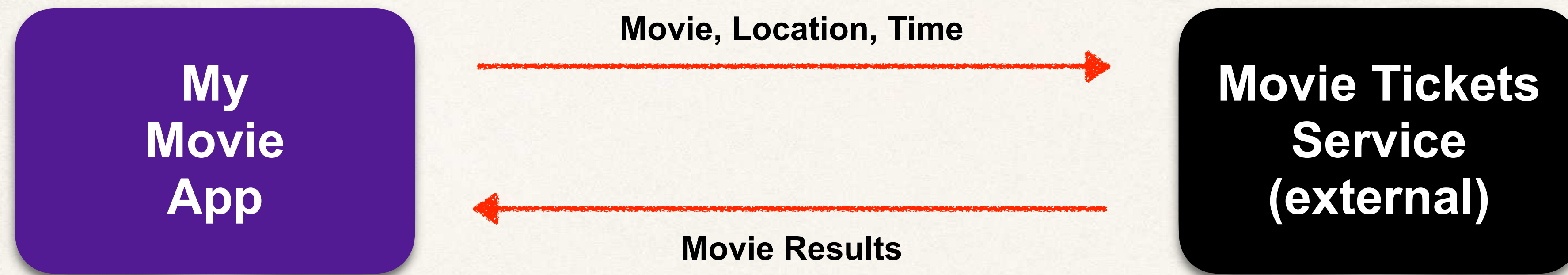
Weather
Service
(external)



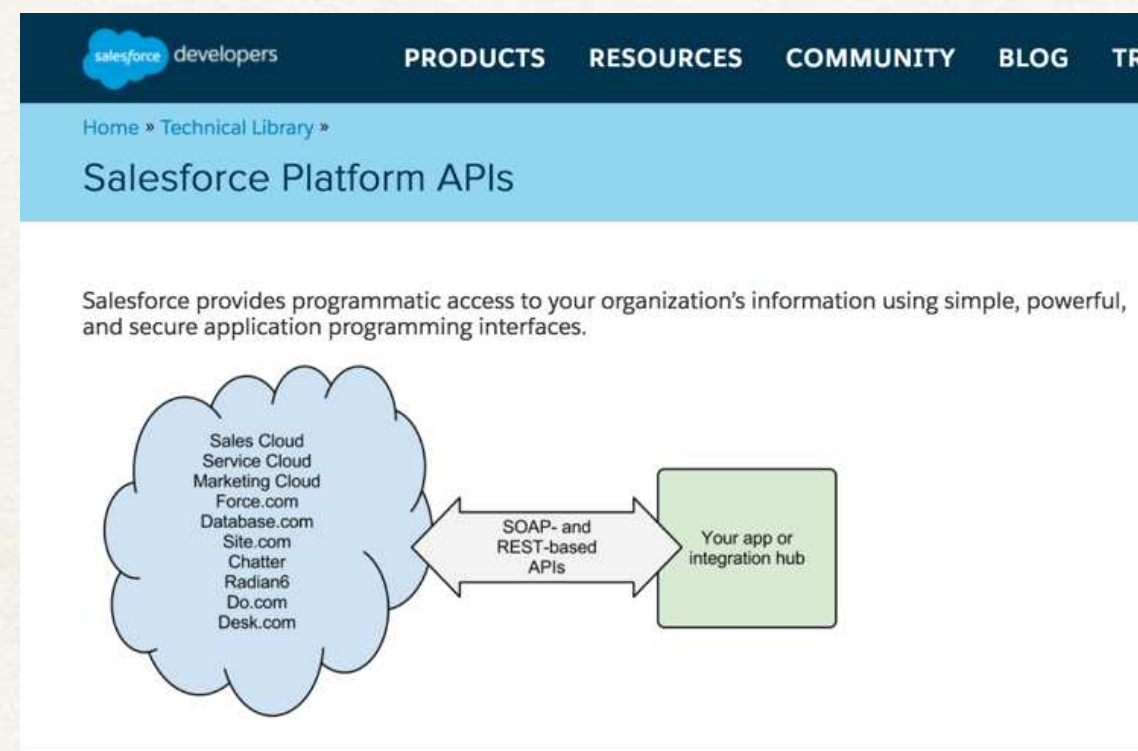
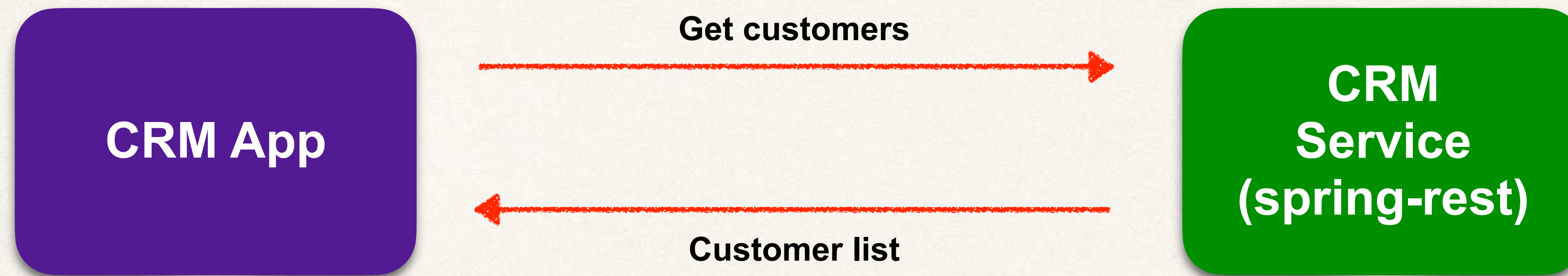
Currency Converter App



Movie Tickets App



Customer Relationship Manager (CRM) App



We will create
this code using
Spring REST
(SERVER)

What do we call it?

REST API

**REST
Web Services**

REST Services

RESTful API

**RESTful
Web Services**

RESTful Services

Generally, all mean the SAME thing

What do we call it?

REST API

**REST
Web Services**

REST Services

RESTful API

**RESTful
Web Services**

RESTful Services

Generally, all mean the SAME thing

JSON Basics - Syntax



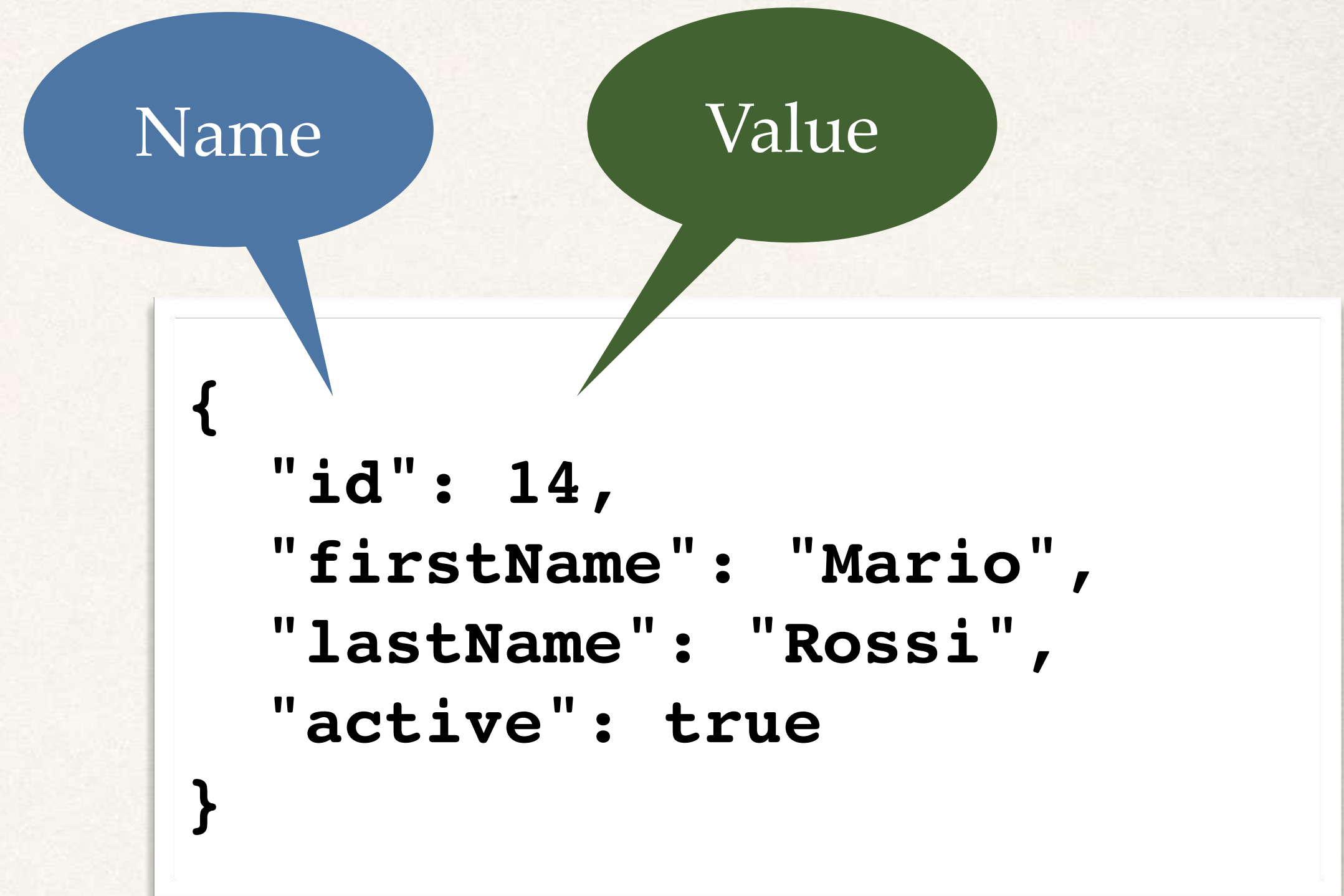
What is JSON?

- JavaScript Object Notation
- Lightweight data format for storing and exchanging data ... plain text
- Language independent ... not just for JavaScript
- Can use with any programming language: Java, C#, Python etc ...

**JSON is just
plain text
data**

Simple JSON Example

- Curley braces define objects in JSON
- Object members are name / value pairs
 - Delimited by colons
- Name is **always** in double-quotes



JSON Values

- Numbers: no quotes
- String: in double quotes
- Boolean: **true**, **false**
- Nested JSON object
- Array
- **null**



Value

```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true,  
  "courses" : null  
}
```


Nested JSON Objects

```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true,  
  "address" : {  
    "street" : "100 Main St",  
    "city" : "Philadelphia",  
    "state" : "Pennsylvania",  
    "zip" : "19103",  
    "country" : "USA"  
  }  
}
```



JSON Arrays

```
{  
  "id": 14,  
  "firstName": "Mario",  
  "lastName": "Rossi",  
  "active": true,  
  "languages" : ["Java", "C#", "Python", "Javascript"]  
}
```



REST HTTP Basics



REST over HTTP

- Most common use of REST is over HTTP
- Leverage HTTP methods for CRUD operations

HTTP Method	CRUD Operation
POST	<u>C</u> reate a new entity
GET	<u>R</u> ead a list of entities or single entity
PUT	<u>U</u> pdate an existing entity
DELETE	<u>D</u> eleate an existing entity

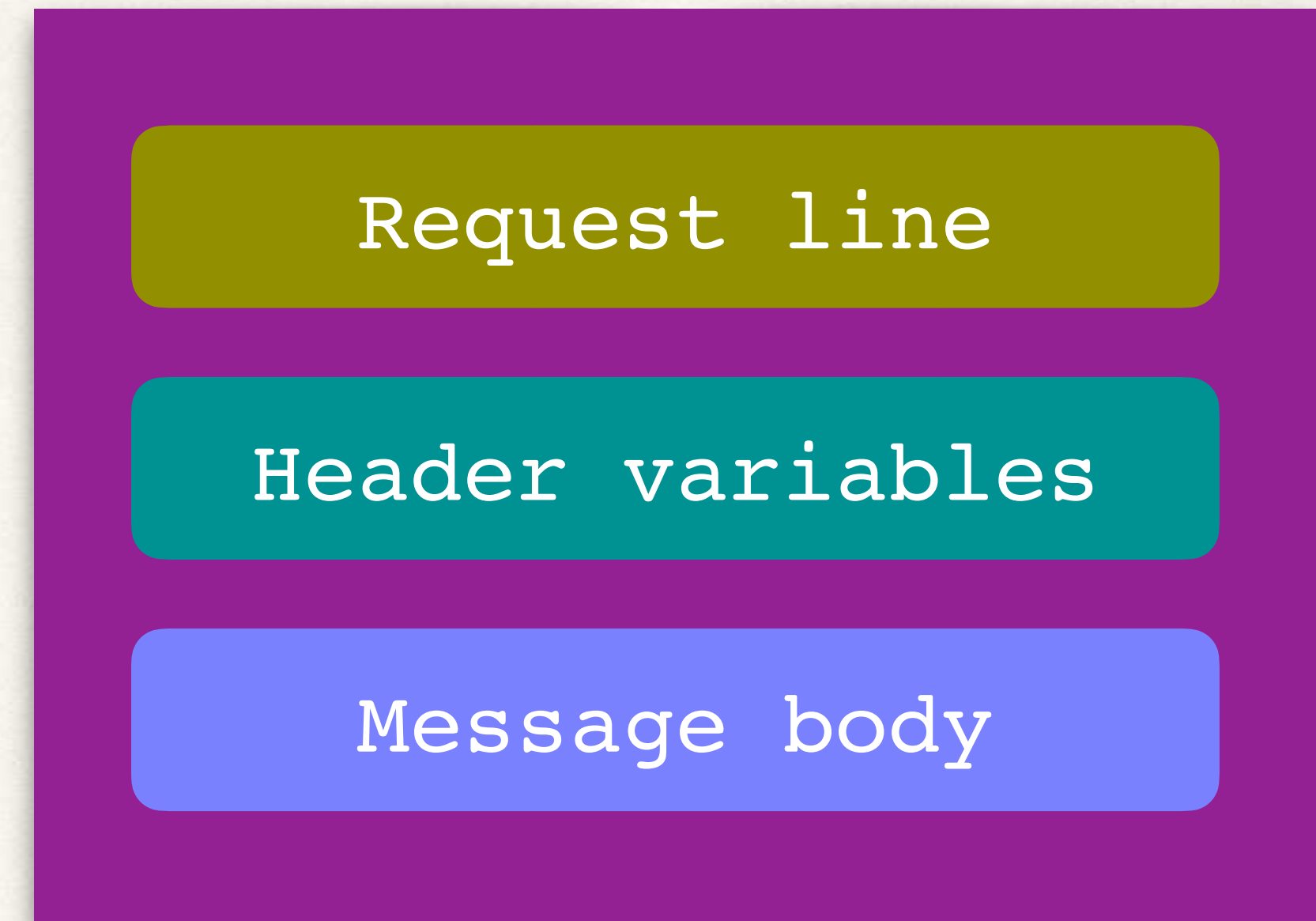
HTTP Messages



HTTP Request Message

- Request line: the HTTP command
- Header variables: request metadata
- Message body: contents of message

HTTP Request Message



HTTP Response Message

- Response line: server protocol and status code
- Header variables: response metadata
- Message body: contents of message

HTTP Response Message

Response line

Header variables

Message body

HTTP Response - Status Codes

Code Range	Description
100 - 199	Informational
200 - 299	Successful
300 - 399	Redirection
400 - 499	Client error
500 - 599	Server error

401 Authentication Required
404 File Not Found

500 Internal Server Error

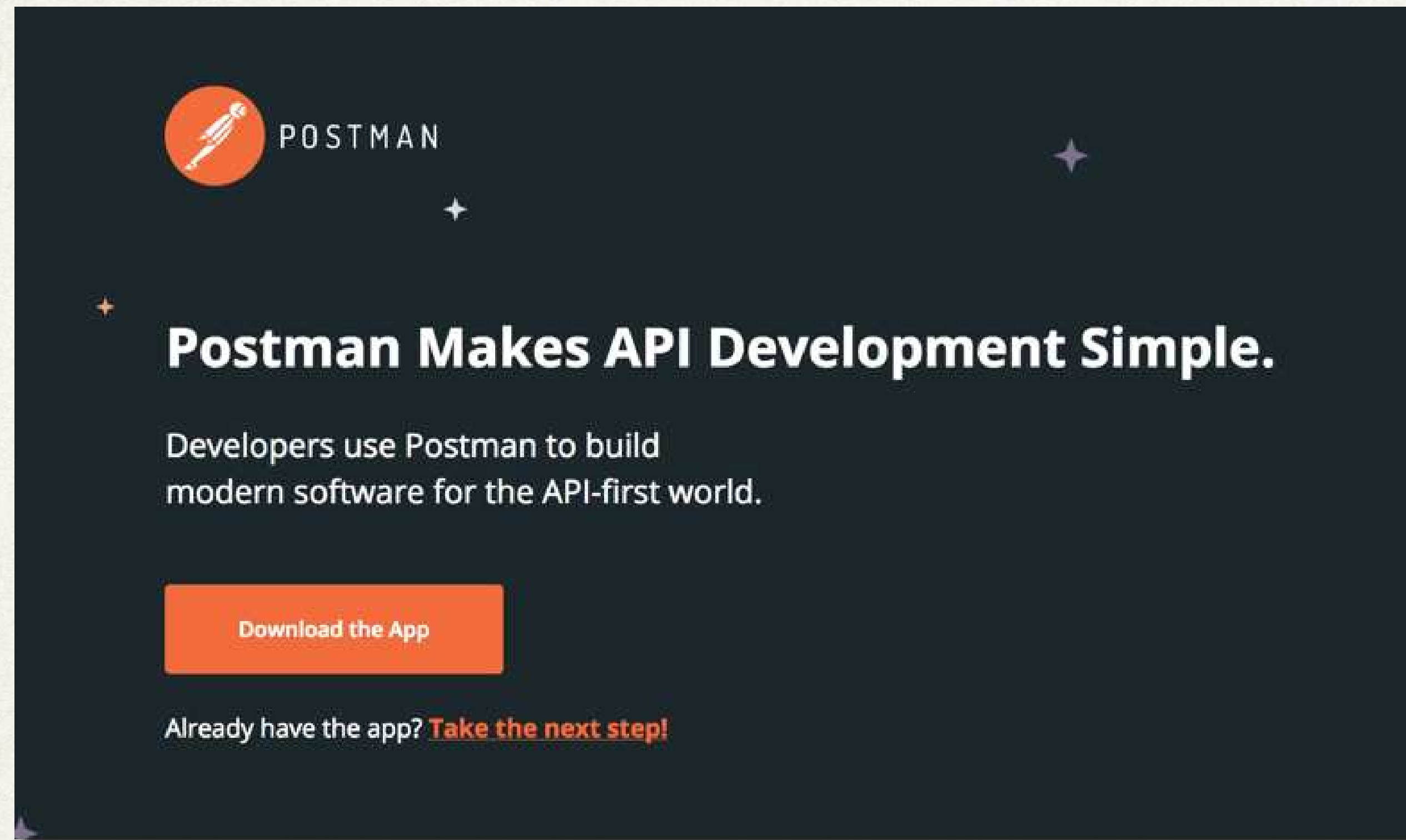
MIME Content Types

- The message format is described by MIME content type
 - Multipurpose Internet Mail-Extension
- Basic Syntax: `type/sub-type`
- Examples
 - `text/html`, `text/plain`
 - `application/json`, `application/xml`, ...

Client Tool

- We need a client tool
- Send HTTP requests to the REST Web Service / API
- Plenty of tools available: curl, Postman, etc ...
- Can also install our own within the application: Swagger

Postman

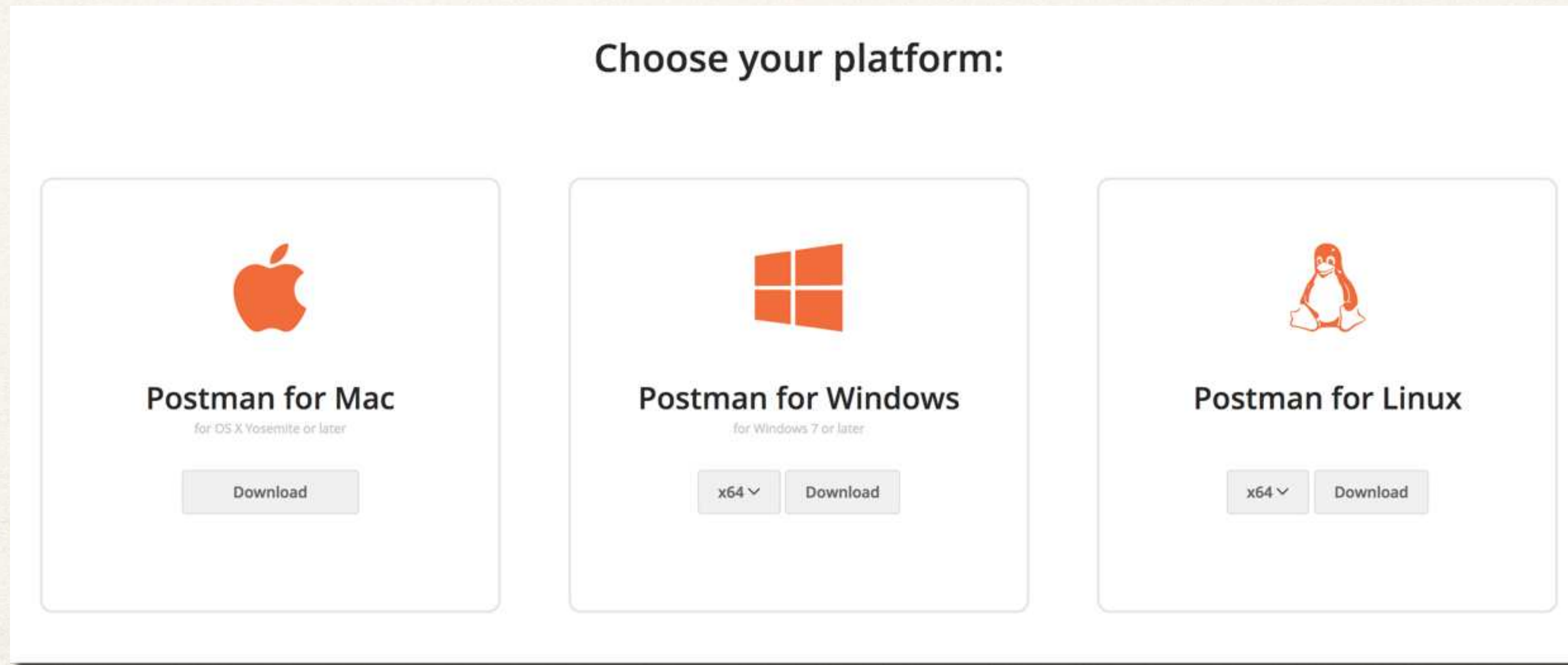


Free
developer
plan

www.getpostman.com

Install Postman is One Option

www.getpostman.com



Setup Swagger Option 2

- We will setup Swagger in code
- Send HTTP requests to the REST Web Service / API
- Allows us to be able to run API endpoints from our application

